



REPORT

AIoT Mangosteen Ripeness Detection

Submitted to

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COE67-312 AI in Embedded Systems

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Abstract

This project implements an edge-AI prototype that classifies mangosteen ripeness as unripe, ripe, or overripe using an ESP32-S3 camera board. From 237 original images, 20 ambiguous samples were excluded, producing a 217-image dataset split into train, validation, and test sets. A compact 96×96 RGB convolutional neural network with 65,075 parameters was trained in Google Colab and constrained below the course limit of 100,000 parameters. The Float32 model achieved 97.22% overall test accuracy and 91.67% balanced test accuracy on 36 held-out images; one overripe sample was predicted as ripe. Full INT8 conversion achieved 100.00% on the same Colab test set. The model was deployed with TensorFlow Lite Micro on a LilyGO T-SIMCAM (ESP32-S3 + OV2640), connected to a web dashboard that freezes each captured image while showing the decision. A preliminary dashboard reading reports 1,281 ms AI inference and 1,577 ms capture-to-result. A full on-device test set, repeated latency percentiles, and INA219 energy measurements were not measured in the supplied hardware evidence and are therefore not claimed.

Project compliance at a glance

Mini-project requirement	Evidence in this report	Status
Three classes on ESP32-S3 camera	Unripe / Ripe / Overripe; first on-device inference and dashboard shown.	Complete
CNN $\leq$ 100,000 parameters	65,075-parameter CNN; architecture in Section 4.	Complete
Float32 validation $\geq$ 88%	100.00% validation accuracy and balanced accuracy (n = 30).	Complete
Full INT8 with $<2\%$ loss	INT8 test score 100.00% vs 97.22% Float32; -2.78 pp change.	Complete
On-device test set + confusion matrix	No full board-side pass was recorded in the supplied evidence.	Not run
Median/P95 latency and INA219 energy	Only single dashboard readings are available; no INA219 log was supplied.	Not run

Report map

- 1. Introduction and related work
- 2. Dataset foundation and label limitations
- 3. Model architecture and training evidence
- 4. Quantization and Colab evaluation
- 5. Hardware deployment and dashboard
- 6. Benchmark card and failure modes
- 7. Group comparison, conclusion, and reproducibility
- Appendix: measurement status and references

The report is intentionally evidence-first: unavailable quantities are marked “Not measured” instead of being estimated.

1. Introduction

Mangosteen quality sorting is normally performed by visual inspection of peel color and surface condition. This creates a useful embedded-AI problem because the system must combine an inexpensive camera, a small neural network, and limited memory while producing a decision understandable to an operator. The project goal is one classification per fruit rather than continuous frame-by-frame sorting: the operator previews a live stream, captures one frame, reviews a frozen image with the result, then starts the next fruit.

The prototype has two engineering objectives. First, it must retain adequate discriminative information at 96×96 RGB without exceeding the 100,000-parameter constraint. Second, its Colab preprocessing must match the board pipeline: class order is [unripe, ripe, overripe], the input is $96 \times 96 \times 3$, and the deployed model is fully INT8. These details are essential because a high notebook score is not meaningful if the embedded input tensor is formed differently.

2. Related Work

Compact convolutional networks are widely used in mobile vision because depthwise/separable feature extraction can reduce computation while retaining useful visual representations [1]. The present work uses a deliberately smaller CNN rather than a large pre-trained model so that the parameter budget, tensor arena, and input path remain feasible on an ESP32-S3. Post-training integer quantization is a practical deployment technique because it converts weights and activations to integer arithmetic compatible with embedded inference [2].

TensorFlow Lite Micro provides an embedded runtime that operates from a fixed tensor arena, so memory allocation and image preprocessing must be designed alongside the model [3]. The ESP32-S3 is suitable for this prototype because it can interface with an OV2640 camera and external PSRAM; however, available RAM and network conditions are runtime constraints, not just board specifications [4]. The course mini-project explicitly evaluates this complete chain: data documentation, model reasoning, deployment, benchmark rigor, and honest failures [5].

Research questions

Question	How this report answers it
Can a $\leq 100k$ CNN separate three ripeness classes?	Yes in Colab: Float32 test accuracy 97.22%; balanced accuracy 91.67% (n = 36).
Does full INT8 conversion retain performance?	Yes on the same Colab test set: 100.00%; small test size is discussed as a limitation.
Can the model run on the target board with a usable workflow?	Yes: first inference, web dashboard, and frozen-frame review are implemented; full board-side evaluation remains pending.

3. Dataset Foundation

The supplied Colab notebook identifies the dataset as Mangosteen Dataset v4. It begins with 237 original images. Twenty images with ambiguous color or unclear labeling were excluded before model selection, leaving 217 usable images. Frames from the same fruit should remain in the same split to avoid leakage; the supplied split records are summarized below.

Split	Unripe	Ripe	Overripe	Total	Use
Train	74	58	19	151	Fitting / augmentation
Validation	15	11	4	30	Model selection
Test	20	12	4	36	Held-out final Colab score
Total	109	81	27	217	After excluding 20 ambiguous images

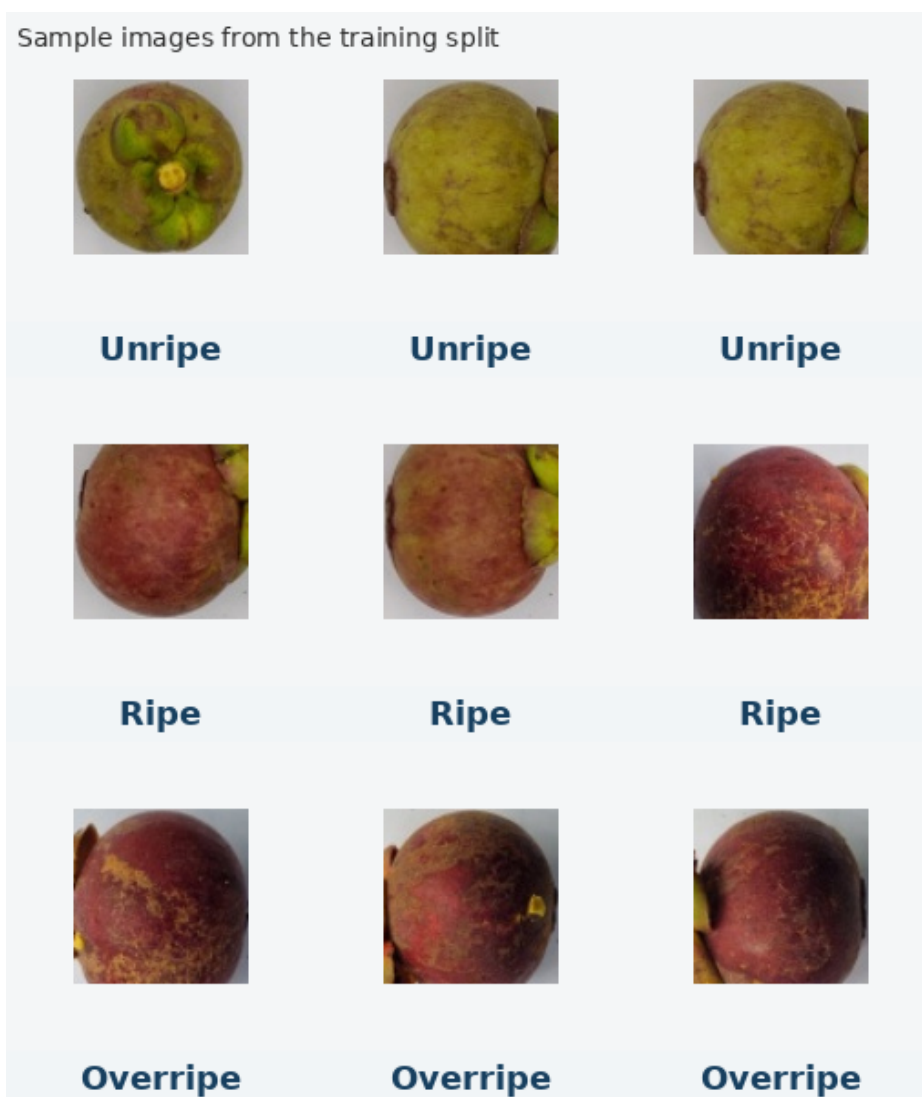


Figure 1. Supplied representative training samples. The three classes use the fixed order unripe, ripe, overripe.

Label quality and limitations

The exclusion of 20 ambiguous samples is a useful quality-control step, but a formal inter-annotator agreement statistic (for example, percent agreement or Cohen’s **K**) was not stored with the supplied files. Therefore, this report does not claim a measured label-agreement value. The override class is also the smallest class (27 images overall and four test images), so its recall has a large uncertainty. This limitation directly explains why balanced accuracy is reported beside overall accuracy.

4. Model Architecture and Training

The model is a compact color-aware CNN named `mangosteen_color_cnn_v3`. It accepts $96 \times 96 \times 3$ RGB images. A coarse color branch uses 8×8 average pooling and flattening, while the feature branch applies three convolution stages followed by a separable convolution. Their features are concatenated before a 64-unit dense layer and the three-class softmax output. The color branch was chosen because peel color is a prominent cue for ripeness; the feature branch contributes texture and spatial structure.

Block	Configuration	Rationale
Input + scaling	$96 \times 96 \times 3$; Rescaling $1/255$	Matches board input contract and normalizes pixels.
Color branch	AveragePool $8 \times 8 \rightarrow$ Flatten	Preserves coarse color information at low cost.
Feature branch	Conv 16 $\rightarrow$ Conv 32 $\rightarrow$ Conv 64; BN + ReLU + MaxPool	Extracts progressively richer visual features.
Compact stage	SeparableConv2D 96; BN + ReLU; GAP	Adds capacity with lower parameter cost.
Classifier	Concatenate $\rightarrow$ Dense 64 ReLU $\rightarrow$ Dropout 0.25 $\rightarrow$ Softmax 3	Fuses color/features and limits overfitting.
Model size	65,075 parameters	Within the $\leq 100,000$ course constraint.

Training protocol

The Colab notebook uses batch size 16, Adam with learning rate 5×10^{-4} , up to 120 epochs, and horizontal/vertical flips, rotation, and zoom applied only to training data. To reduce the impact of the smaller overripe class, the training stream samples class IDs with weights 0.30 / 0.30 / 0.40. Three random seeds (11, 42, 73) are evaluated; the saved candidate with the best validation balanced accuracy is selected. ModelCheckpoint, early stopping (patience 25), and ReduceLROnPlateau (patience 7, factor 0.5) are used to preserve the lowest-validation-loss candidate.

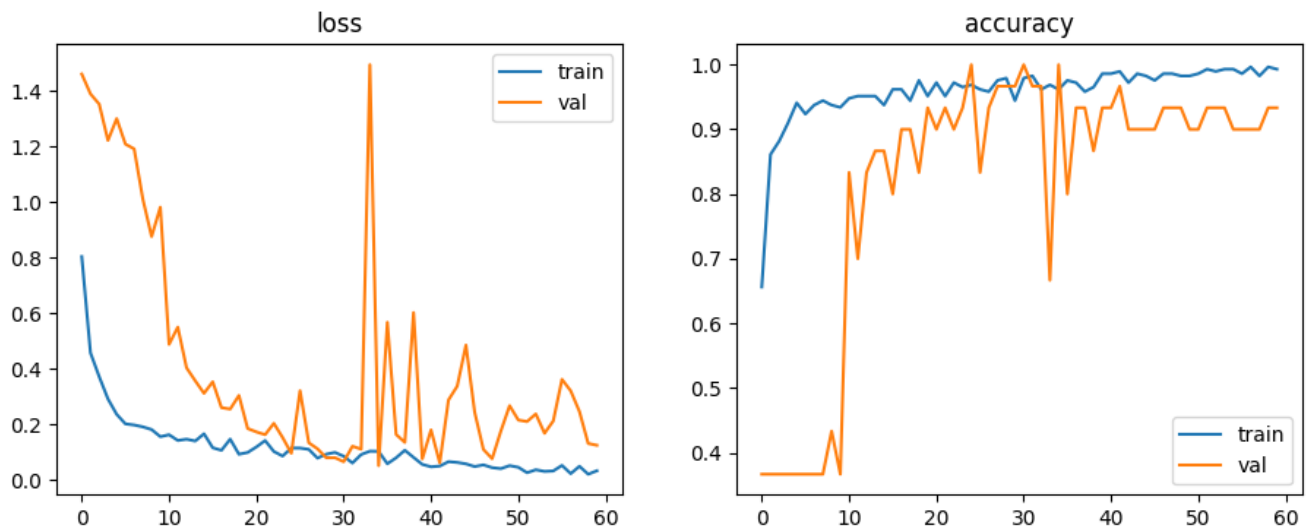


Figure 2. Actual training and validation loss/accuracy history exported by the supplied Colab notebook for the selected run.

5. Quantization and Colab Evaluation

The chosen Keras model is converted using TensorFlow Lite post-training full integer quantization. The converter is supplied with 120 representative training images, restricts operations to INT8 built-ins, and sets both the input and output tensor types to int8. During evaluation, input pixels are quantized using the interpreter's scale and zero point before invocation. This is the contract that the firmware must reproduce.

Metric	Float32	Full INT8	Interpretation
Validation accuracy (n = 30)	100.00%	—	Validation score for selected Float32 model.
Validation balanced accuracy	100.00%	—	All validation classes correctly predicted.
Test accuracy (n = 36)	97.22%	100.00%	INT8 is +2.78 pp on this small test set.
Test balanced accuracy	91.67%	100.00%	Float32 exposes weaker overripe recall.
Float32 errors	1 overripe → ripe	0	Use the confusion matrix, not accuracy alone.

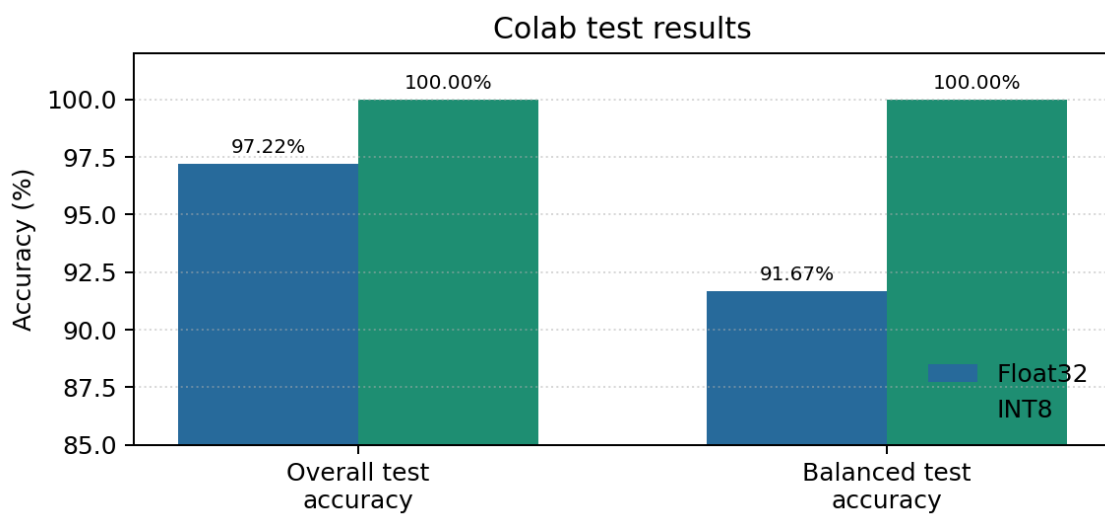


Figure 3. Colab test summary from the supplied report artifacts. The INT8 increase must be interpreted cautiously because the held-out set has only 36 images.

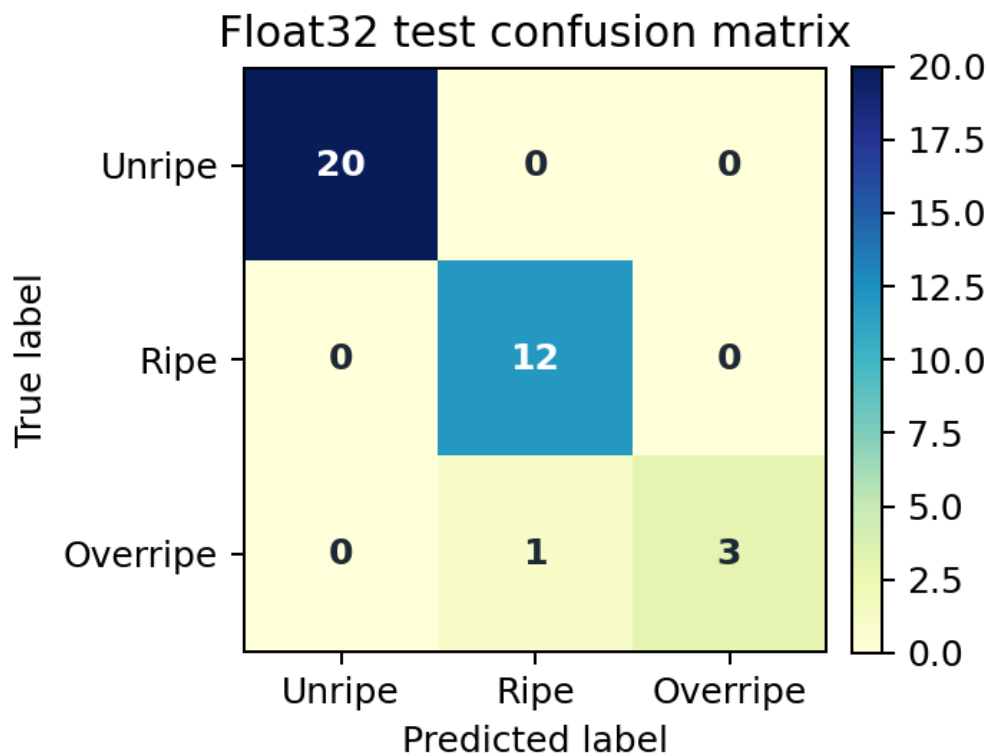


Figure 4. Float32 held-out test confusion matrix: 35/36 correct, with one overripe sample predicted as ripe.

A “negative drop” after quantization does not prove that INT8 is universally better; it only describes this one small test set. The final system evaluation should re-run the exact held-out set through the board and record a board-side confusion matrix.

6. Hardware Deployment

The target platform is a LilyGO T-SIMCAM based on ESP32-S3 with an OV2640 camera and PSRAM. TensorFlow Lite Micro runs the INT8 model from a fixed tensor arena. The firmware is responsible for camera acquisition, JPEG decode and resize to 96×96 RGB, conversion to the INT8 tensor format, inference, and HTTP response generation. The configured tensor arena is 2.0 MiB in PSRAM; peak arena usage has not yet been instrumented and is not inferred from free-PSRAM readings.

Component	Implementation in this project	Evidence / caveat
Camera	OV2640; 320 × 240 JPEG capture before model preprocessing	Camera initializes successfully in serial output.
Controller	LilyGO T-SIMCAM / ESP32-S3	Runs TensorFlow Lite Micro firmware.
Model input	96 × 96 × 3 RGB, class order: unripe, ripe, overripe	Must remain identical to Colab order and preprocessing.
Runtime memory	2.0 MiB tensor arena in PSRAM	Configured allocation; peak use pending.
Network UI	Local Wi-Fi dashboard	AI continues without page if Wi-Fi setup fails.

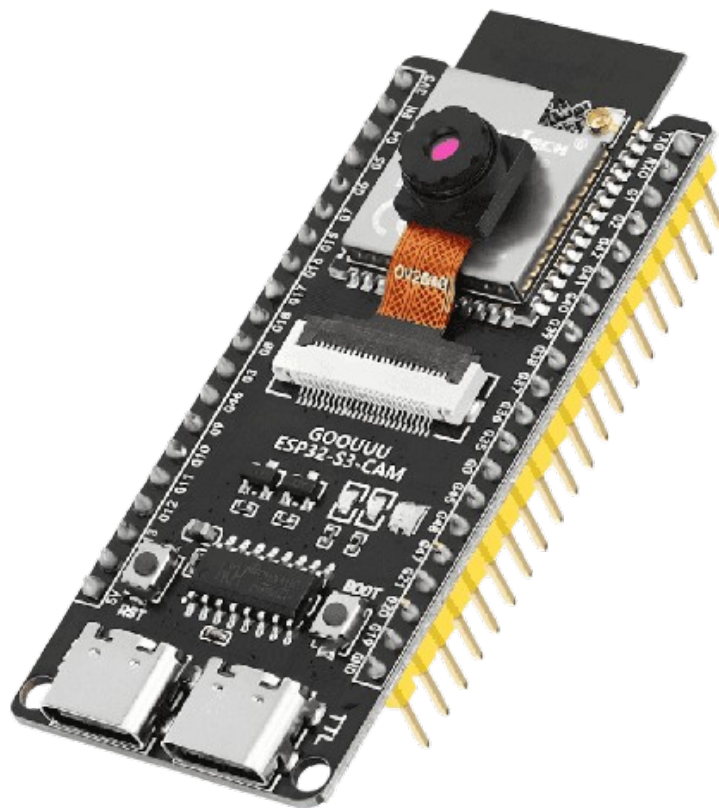


Figure 5. LilyGO T-SIMCAM / ESP32-S3 camera board used for deployment (project presentation source).

Firmware pipeline

Stage	Firmware action	Output
1. Preview	Serve a low-latency camera stream to the browser.	Operator positions one fruit.
2. Capture	Receive one web trigger and acquire a JPEG frame.	A single inspection image.
3. Preprocess	Decode; form 96×96 RGB; apply model INT8 scale and zero point.	Input tensor matching Colab.
4. Infer	Invoke TensorFlow Lite Micro and read three class scores.	Predicted class and confidence.
5. Review	Freeze the captured image, show result and benchmark values; update class total.	One decision per fruit.

First Successful On-Device Inference

The serial-monitor record below is direct hardware-side evidence that the camera and the deployed model initialize successfully. It reports the COM3 connection, free PSRAM, camera initialization, model loading, the $96 \times 96 \times 3$ input contract, INT8 input scale and zero point, three class scores, and the final prediction. This satisfies the Week 2 requirement to show a first successful inference on the target board.

```
--- Terminal on COM3 | 115200 8-N-1
--- Available filters and text transformations: debug, default, direct, esp32_exception_decoder, hexlify,
printable, send_on_enter, time
--- More details at https://bit.ly/pio-monitor-filters
--- Quit: Ctrl+C | Menu: Ctrl+T | Help: Ctrl+T followed by Ctrl+H

Mangosteen AI starting...
Free PSRAM: 8386295 bytes
Camera initialized successfully
Loading mangosteen model...
Model loaded successfully
Input dimensions: 96 x 96 x 3
Input type: 9
Input scale: 1.000000
Input zero point: -128
Tensor Arena remaining
=====
Mangosteen classification
=====
unripe: 0.39%
ripe: 0.00%
overripe: 99.61%
-----
Prediction: overripe
```

Figure 6. Serial Monitor evidence from the deployed ESP32-S3: camera initialized, model loaded, INT8 input configured, and a final prediction of overripe.

This screen proves that the embedded inference pipeline runs. It is not a board-side accuracy test because it shows one captured frame rather than the complete 36-image held-out test set.

7. Web Dashboard and Preliminary Benchmark Card

The dashboard supports the workflow selected for this prototype: live video during positioning, then a single “Capture & Classify” action. After capture, the exact image and its class scores remain frozen for review. Pressing “New Inspection” returns to live preview for the next fruit. This avoids treating a 1.281-second inference as a real-time per-frame video classifier and provides a clear record of one classification per fruit.

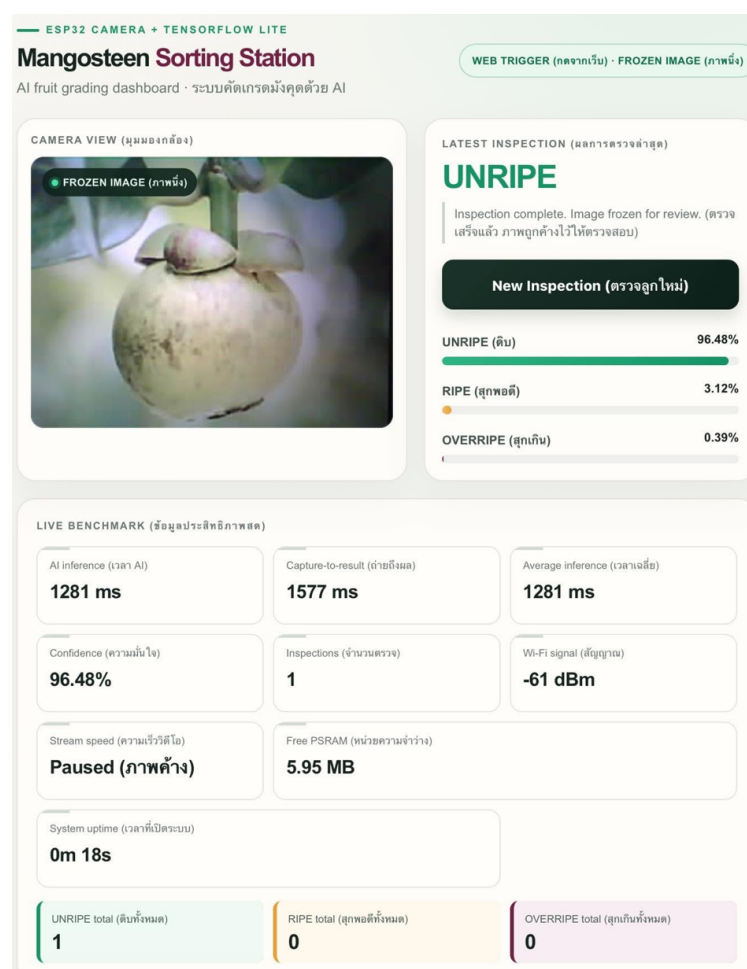


Figure 7. Implemented web dashboard: frozen captured image, English labels with Thai translations, class probabilities, counters, and live benchmark fields.

Benchmark item	Current reading	Status / interpretation
AI inference	1,281 ms	Single dashboard observation; repeat sampling required.
Capture-to-result	1,577 ms	Single dashboard observation; includes capture path.
Derived throughput	0.63 inspections/s	1000 / 1577; not a continuous-video FPS claim.
Free PSRAM	≈ 5.95 MB	Startup/dashboard reading; not measured peak tensor arena use.
On-device test accuracy + CM	Not measured	No complete 36-image board-side log was supplied.
Latency median / P95	Not measured	No repeated-trial log was supplied.
Active / sleep energy	Not measured	No INA219 measurement log was supplied.

Benchmark status: preliminary. The current measurements demonstrate an end-to-end working system, but they are not sufficient to claim final latency distribution, energy, or board-side accuracy.

8. Failure Modes and Lessons Learned

This section records evidence and unresolved work rather than converting incomplete measurements into optimistic claims. It is included because the project handout explicitly rewards honest failure analysis.

Issue or abandoned approach	Evidence / impact	Mitigation or next action
Continuous repeated classification	A single AI inference is about 1.281 s, so classifying every video frame is not a usable live-video experience.	Replaced with operator-triggered capture, frozen result review, and a separate New Inspection action.

Issue or abandoned approach	Evidence / impact	Mitigation or next action
Small, imbalanced overripe class	Only 27 overripe images total and four in the test set; Float32 made one overripe → ripe error.	Collect more overripe fruit across lighting/backgrounds; retain fruit-level split; report balanced metrics.
Wi-Fi connection failure	Serial output reports that AI continues without the web page when Wi-Fi connection fails.	Verify SSID/password, use 2.4 GHz Wi-Fi, then record and open the board IP address.
Board-side evaluation incomplete	Colab test accuracy does not verify the camera/JPEG/INT8 firmware path.	Run the held-out set on target hardware and publish a board-side confusion matrix.
Latency distribution and energy incomplete	Only one displayed latency value is available; no INA219 log is supplied.	Record repeated trials; report median/P95 and active/sleep energy with units and protocol.

What should not be claimed yet

- Do not label 100.00% INT8 Colab accuracy as final hardware accuracy.
- Do not describe the 1,281 ms dashboard value as median latency or P95 latency.
- Do not infer peak tensor-arena use from free PSRAM, or infer energy without the INA219 measurement.
- Do not explain the group performance difference as causal until both teams use the same benchmark protocol.

9. Group Comparison

The supplied presentation includes a side-by-side comparison with the other group. The values below are preserved as presentation-reported results. They are useful for discussion, but they should be re-measured under one agreed capture protocol before they are treated as a causal engineering comparison.

Metric	Our group	Other group	Observed difference	Comparison caution
Test accuracy, Float32	97.22%	75.00%	+22.22 pp	Same-data claim needs shared split verification.
Test accuracy, INT8	100.00%	58.30%	+41.70 pp	Other-group model and preprocessing were not independently audited here.
AI inference	1,281 ms	~1,200 ms	~+81 ms	Approximate, single-result figures.
Capture-to-result	1,577 ms	~1,500 ms	~+77 ms	Approximate, single-result figures.
Free PSRAM	5.95 MB	5.90 MB	+0.05 MB	Free memory is not peak arena use.
Throughput	0.63 inspections/s	0.67 inspections/s	-0.04 inspections/s	Derived from the reported capture-to-result time.

Interpretation

Our group’s supplied Colab scores are higher, while the other group’s approximate timing is slightly faster. The report cannot attribute this difference to one architecture choice because the model, preprocessing, firmware, measurement repetition, and environmental conditions were not standardized in the supplied evidence. The appropriate conclusion is that the systems differ and should be compared again using the same fruit set, board settings, Wi-Fi state, and timing protocol.

10. Conclusion and Future Work

This project demonstrates a complete edge-AI path from a cleaned mangosteen dataset to a compact CNN, full INT8 conversion, first on-device inference, and a practical frozen-frame web workflow. The 65,075-parameter model satisfies the embedded constraint and gives strong Colab test results. The dashboard makes the interaction understandable for manual sorting: each fruit has one image, one result, and one counter update.

The most important remaining work is measurement, not another optimistic model claim. The team should complete a board-side held-out test, repeated timing (median and P95), and INA219 active/sleep energy logs. These additions will turn the current preliminary benchmark card into the final evidence required by the course.

Three concrete extensions

Priority	Extension	Success criterion
1	Complete hardware-side evaluation	36 held-out images, board-side accuracy and confusion matrix.
2	Reduce capture-to-result time	Profile JPEG decode, resize, and inference; target <500 ms only after baseline repeats.
3	Automate physical sorting	Add a sensor trigger and servo/conveyor decision after a confirmed single capture.

Reproducibility

The supplied Google Colab notebook contains dataset loading, the exact class order, architecture, training schedule, test evaluation, full INT8 conversion, exported metrics, and training history. Use the notebook below as the reproducible source for model-side results: [Open the supplied Colab notebook](#) Before running it again, keep the final dataset ZIP and the exported results ZIP together. Archive class_names.json, metrics.json, training_history.csv, and the exact .tflite model used to generate model_data.h.

Appendix A. Measurement Status

This table records the measurement state supported by the supplied hardware evidence. No metric below is extrapolated from a single image or estimated without a device log.

Metric	Evidence available now	Reported status
On-device accuracy	One successful serial-monitor inference is shown; no 36-image board-side log was supplied.	Not measured
Inference latency	Dashboard shows one 1,281 ms AI-inference reading.	Preliminary single reading; median/P95 not measured
Capture-to-result	Dashboard shows one 1,577 ms capture-to-result reading.	Preliminary single reading; median/P95 not measured
Energy	No INA219 active/sleep log was supplied.	Not measured
Tensor arena	Firmware configures a 2.0 MiB PSRAM tensor arena; no peak-use log was supplied.	Configured size 2.0 MiB; peak use not measured

Appendix B. References

[1] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, “MobileNetV2: Inverted Residuals and Linear Bottlenecks,” Proc. CVPR, 2018.

[2] TensorFlow, “Post-training integer quantization,” TensorFlow Model Optimization documentation, accessed Sep. 2026.

[3] Google AI Edge, “LiteRT for Microcontrollers,” documentation, accessed Sep. 2026.

[4] Espressif Systems, “ESP32-S3 Series Datasheet,” version 2.2, accessed Sep. 2026.

[5] COE67-312 AI in Embedded Systems, “AIoT Mangosteen Ripeness Detection: Mini Project,” course handout, 2026.

[6] Project group, “Mangosteen Dataset v4 — Balanced augmentation > INT8,” supplied Google Colab notebook, 2026.

Source figures: Figures 1, 3, 4, and 7 are supplied project report artifacts; Figure 2 is exported from the supplied Colab notebook; Figure 5 is cropped from the supplied project presentation; Figure 6 is the supplied Serial Monitor screenshot.